

POLICY FRAMEWORK

Anti-Money Laundering Transaction Monitoring & Compliance Rules

Based on IBM Research Synthetic AML Dataset | CDLA-Sharing-1.0

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This document defines enforceable compliance rules for automated detection of anti-money laundering (AML) violations within financial transaction databases. Rules are ingested by the PolicyPilot compliance agent and applied during scheduled and on-demand scans.

SECTION 1 — POLICY OVERVIEW & REGULATORY SCOPE

1.1 Purpose

This policy framework establishes machine-enforceable rules for the detection and flagging of suspicious financial transactions within any database connected to the PolicyPilot compliance agent. Rules are derived from established AML regulatory frameworks including the Bank Secrecy Act (BSA), FATF Recommendations, EU AMLD6, and FinCEN guidance, adapted for the IBM synthetic AML transaction dataset structure.

1.2 Dataset Reference

These rules are designed to operate against the IBM Transactions for Anti-Money Laundering (AML) dataset published on Kaggle under CDLA-Sharing-1.0 license. The dataset contains synthetic multi-agent financial transaction records with the following schema:

Field	Type	Description
Transaction ID	String	Unique identifier for each transaction
Timestamp	DateTime	Date and time of transaction execution
From Bank	String	Originating bank/institution code
From Account	String	Source account identifier
To Bank	String	Receiving bank/institution code
To Account	String	Destination account identifier
Amount Received	Decimal	Amount credited to destination (USD)
Receiving Currency	String	Currency of received amount
Amount Paid	Decimal	Amount debited from source (USD)
Payment Currency	String	Currency of paid amount
Payment Format	String	Wire / ACH / Check / Credit Card / Cash / etc.
Is Laundering	Boolean	Ground truth label (0=clean, 1=suspicious)

1.3 Regulatory Basis

- **BSA / FinCEN:** Requires financial institutions to report transactions over \$10,000 (CTR) and maintain 5-year record retention.
- **FATF Rec. 10:** Prohibits structuring — splitting transactions to avoid reporting thresholds (smurfing).
- **EU AMLD6:** Extends AML liability to a broader set of predicate offenses; requires enhanced due diligence for high-risk jurisdictions.
- **OFAC Sanctions:** Prohibits transactions with sanctioned entities, countries, or individuals appearing on SDN lists.
- **PCI-DSS 3.0:** Requires encryption of cardholder data at rest and in transit for all credit card transactions.

1.4 Scope

This policy applies to ALL financial transaction records ingested into any database monitored by the PolicyPilot agent, regardless of transaction type (wire transfer, ACH, credit card, check, or cash equivalent). Rules are evaluated at ingestion time and during periodic compliance scans scheduled at configurable intervals.

SECTION 2 — ENFORCEABLE COMPLIANCE RULES

The following rules are machine-enforceable and will be automatically applied by the PolicyPilot compliance agent during database scans. Each rule specifies a unique Rule ID, risk classification, detection logic, and the remediation action to be triggered.

AML-001 Large Cash Transaction Reporting (CTR)				CRITICAL
Category:	Transaction Threshold	IBM Field(s):	Amount Paid / Amount Received	
Rule:	Any single transaction where Amount Paid or Amount Received exceeds \$10,000 USD (or equivalent in any currency) must be flagged for Currency Transaction Report (CTR) filing within 15 business days.			
Detection Logic:	<code>SELECT * FROM transactions WHERE amount_paid > 10000 OR amount_received > 10000</code>			
Action:	Flag for CTR filing. Block further processing until compliance officer reviews.			

AML-002 Structuring / Smurfing Detection				CRITICAL
Category:	Pattern Detection	IBM Field(s):	From Account, To Account, Timestamp, Amount Paid	
Rule:	Any account that executes 3 or more transactions totaling over \$10,000 to the same destination account within a 24-hour window, where each individual transaction is below \$10,000, must be flagged as potential structuring.			
Detection Logic:	<code>GROUP BY from_account, to_account within 24h; SUM(amount) > 10000 AND COUNT(*) >= 3 AND MAX(amount) < 10000</code>			
Action:	CRITICAL flag. Freeze account pending SAR filing. Notify compliance officer.			

AML-003 High-Frequency Transfer Velocity				HIGH
Category:	Velocity Check	IBM Field(s):	From Account, To Account, Timestamp	
Rule:	Any account executing more than 10 outbound transfers within a 24-hour period, or more than 5 transfers to the same beneficiary account within 24 hours, must be flagged for enhanced due diligence review.			
Detection Logic:	<code>GROUP BY from_account WHERE COUNT(*) > 10 in 24h OR (COUNT(*) > 5 AND same to_account)</code>			
Action:	HIGH flag. Require EDD documentation. Temporarily restrict account to < \$1,000 single-transaction limit.			

AML-004 Cross-Currency Conversion Monitoring				HIGH
Category:	Currency Risk	IBM Field(s):	Payment Currency, Receiving Currency, Amount Received	
Rule:	Transactions where the Payment Currency differs from the Receiving Currency AND the converted amount exceeds \$5,000 USD equivalent must be flagged. Foreign exchange transactions from high-risk jurisdictions require immediate review.			
Detection Logic:	WHERE payment_currency != receiving_currency AND amount_received_usd > 5000			
Action:	HIGH flag. Verify exchange rate legitimacy. Check counterparty jurisdiction against FATF grey/black lists.			
AML-005 Round-Amount Transaction Clustering				MEDIUM
Category:	Pattern Detection	IBM Field(s):	Amount Paid, Timestamp	
Rule:	Accounts that consistently send exactly round-dollar amounts (e.g., \$1,000, \$5,000, \$9,999) across multiple transactions exhibit laundering behavior. Flag any account where > 70% of transactions in a 7-day window are round-dollar amounts.			
Detection Logic:	WHERE amount % 1000 = 0 OR amount = 9999 OR amount = 4999; grouped frequency analysis			
Action:	MEDIUM flag. Add to enhanced monitoring list. Require source-of-funds documentation.			
AML-006 Unknown Payment Format Restriction				MEDIUM
Category:	Data Quality	IBM Field(s):	Payment Format	
Rule:	Transactions with a Payment Format value not in the approved set [Wire, ACH, Cheque, Credit Card, Cash, Reinvestment, Bitcoin] must be flagged as data quality violations and held for manual review.			
Detection Logic:	WHERE payment_format NOT IN ('Wire', 'ACH', 'Cheque', 'Credit Card', 'Cash', 'Reinvestment', 'Bitcoin')			
Action:	MEDIUM flag. Reject transaction from processing. Request format clarification from originating institution.			
AML-007 Same-Bank Internal Cycling				HIGH
Category:	Pattern Detection	IBM Field(s):	From Bank, To Bank, From Account, To Account, Timestamp	
Rule:	Transactions where From Bank equals To Bank AND the funds cycle back to the originating account (or a closely related account) within 72 hours suggest layering. Flag any such circular flow exceeding \$2,000.			
Detection Logic:	WHERE from_bank = to_bank; trace circular flows within 72h returning to origin account cluster			
Action:	HIGH flag. Block transaction. Submit SAR within 30 days if laundering confirmed.			

AML-008 Suspicious Beneficiary Concentration				HIGH
Category:	Network Analysis	IBM Field(s):	To Account, From Account, Timestamp	
Rule:	Any single destination account that receives funds from more than 5 distinct source accounts within a 48-hour period must be flagged as a potential money mule recipient account.			
Detection Logic:	GROUP BY to_account WHERE COUNT(DISTINCT from_account) > 5 within 48h			
Action:	HIGH flag. Escalate to financial crime investigation unit. Enhanced monitoring for 90 days.			
AML-009 Rapid Succession Micro-Transactions				MEDIUM
Category:	Pattern Detection	IBM Field(s):	From Account, Amount Paid, Timestamp	
Rule:	Any account executing more than 20 transactions each under \$200 within a 1-hour window (characteristic of automated layering systems) must be flagged.			
Detection Logic:	GROUP BY from_account in 1h window WHERE COUNT(*) > 20 AND MAX(amount) < 200			
Action:	MEDIUM flag. Temporarily suspend account API access. Manual review required.			
AML-010 Dormant Account Sudden Activation				CRITICAL
Category:	Account Behavior	IBM Field(s):	From Account, Timestamp, Amount Paid	
Rule:	Any account with no recorded transaction in the preceding 180 days that suddenly executes a transaction exceeding \$2,500 must be flagged for identity verification and source-of-funds check.			
Detection Logic:	Identify accounts with last_transaction_date > 180 days ago; flag on new transaction if amount > 2500			
Action:	CRITICAL flag. Require re-KYC verification. Hold funds for 5 business days pending review.			

SECTION 3 — VIOLATION SEVERITY, REPORTING & ENFORCEMENT

3.1 Severity Classification Matrix

Severity	Rule(s)	Auto-Action	Human Review SLA	Regulatory Report
CRITICAL	AML-001, AML-002, AML-010	Transaction held + account flagged	< 4 hours	CTR / SAR required
HIGH	AML-003, AML-004, AML-007, AML-008	Enhanced monitoring activated	< 24 hours	SAR if confirmed
MEDIUM	AML-005, AML-006, AML-009	Flag for review + restrict limits	< 72 hours	Internal report only
LOW	All others	Log and monitor	< 7 business days	None required

3.2 Automated Reporting Requirements

Currency Transaction Report (CTR): Filed for any transaction > \$10,000. Must be submitted to FinCEN within 15 calendar days.

Suspicious Activity Report (SAR): Filed when a pattern suggests money laundering or structuring. Submit within 30 days of detection; 60 days if no suspect identified.

Internal Compliance Log: All flagged transactions must be logged in the audit trail with: timestamp, rule ID, amount, account IDs, and analyst action taken.

Enhanced Due Diligence (EDD) Record: For all HIGH and CRITICAL violations: document source of funds, account ownership verification, and final disposition within 30 days.

3.3 Data Retention for Compliance Records

All transaction records, violation flags, CTRs, SARs, and EDD documentation must be retained for a minimum of 5 years from the date of the transaction per BSA/FinCEN requirements. PolicyPilot auto-schedules retention reviews 90 days before any record reaches its deletion window.

3.4 Policy Enforcement Integration

These rules are loaded into PolicyPilot as machine-executable compliance policies. The compliance agent scans connected databases on configurable intervals, applies all AML rules in this document, and produces: (a) a real-time violations dashboard, (b) per-database compliance scores, (c) auto-generated remediation SQL/NoSQL queries, and (d) an audit-ready export in CSV, JSON, or PDF format.

3.5 Document Approval

Role	Name	Signature	Date
Chief Compliance Officer	Parth Bhatt	_____	Feb 2026

Head of Financial Crime	Compliance Team	_____	Feb 2026
PolicyPilot Admin	GDG Hackfest Team	_____	Feb 2026

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